Random Stackings and Stacking Faults

The Hexagonality Law at work: colloids, ice and the faults of crystals

DICTO (artist, Osaka, Japan) · contact@dictospheres.com · https://dictospheres.com
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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Abstract. The Hexagonality Law (#66) says that in a stacking of close-packed layers, where space is shared out by weight among atoms and holes, each kind of piece totals exactly what its layers hold in their own pure stackings. DICTO proposed to look for the conditions where this matters in real matter. Two follow at once, and both are exact. **Theorem 1 (any stacking):** however the layers are stacked, repeating, faulted or random, with any correlations, each kind of piece averages exactly V(cc) + h·(V(hh) − V(cc)) per layer, where h is the share of hexagonal layers; so the single number that diffraction measures in randomly stacked colloids and in stacking-disordered ice fixes every hole volume exactly. **Theorem 2 (faults):** a stacking fault changes the total of every kind of piece by exactly the number of layers it turns hexagonal (or cubic) times V(hh) − V(cc); the classical faults of cubic crystals (twin, intrinsic, extrinsic) turn 1, 2 and 2 layers hexagonal, those of hexagonal crystals (I1, I2, E) turn 1, 2 and 3 layers cubic. In rock salt the balls gain at each fault exactly what the holes lose. The one piece the law's proof does not cover, the Laves foot atom that reads three layers, keeps the law exactly on every stacking up to eight layers (390 of 390 exact tests).

1. Where the law matters

Paper #66 proved the Hexagonality Law for every kind of piece fixed by its two neighbouring layers, and checked it exactly in 1064 cases across pyrochlore, rock salt, cadmium iodide, zinc blende and the Laves phases. DICTO proposed the next step: to find the conditions in real matter where the law, the Swap Rule and the rule of halves play a significant part. Two such conditions are everywhere in crystals: stackings that are not periodic at all, and faults in otherwise perfect stackings. For both the law gives exact statements with no further computation, and this paper sets them out.

2. Setting

As in #57, #64 and #66: close-packed layers in positions A, B, C; a layer is cubic (c) when its two neighbours differ and hexagonal (h) when they are the same (Jagodzinski); the sites of each gap (a ball site, two tetrahedral sites, an octahedral site) share space by weight in a power diagram, atoms with weight λ, empty holes 0. A *role* is a kind of site, and the law applies to every role whose piece in the gap between layers of types x and y has a volume F(x, y) fixed by those two types; #64 found this for every role computed except the Laves foot atom above λ = 17/96. For such a role, #66 proved that the Swap Rule F(c, h) + F(h, c) = F(c, c) + F(h, h) holds exactly when F(x, y) = f(x) + g(y), so that over any closed run of layers the total is $n_c F(c, c) + n_h F(h, h)$.

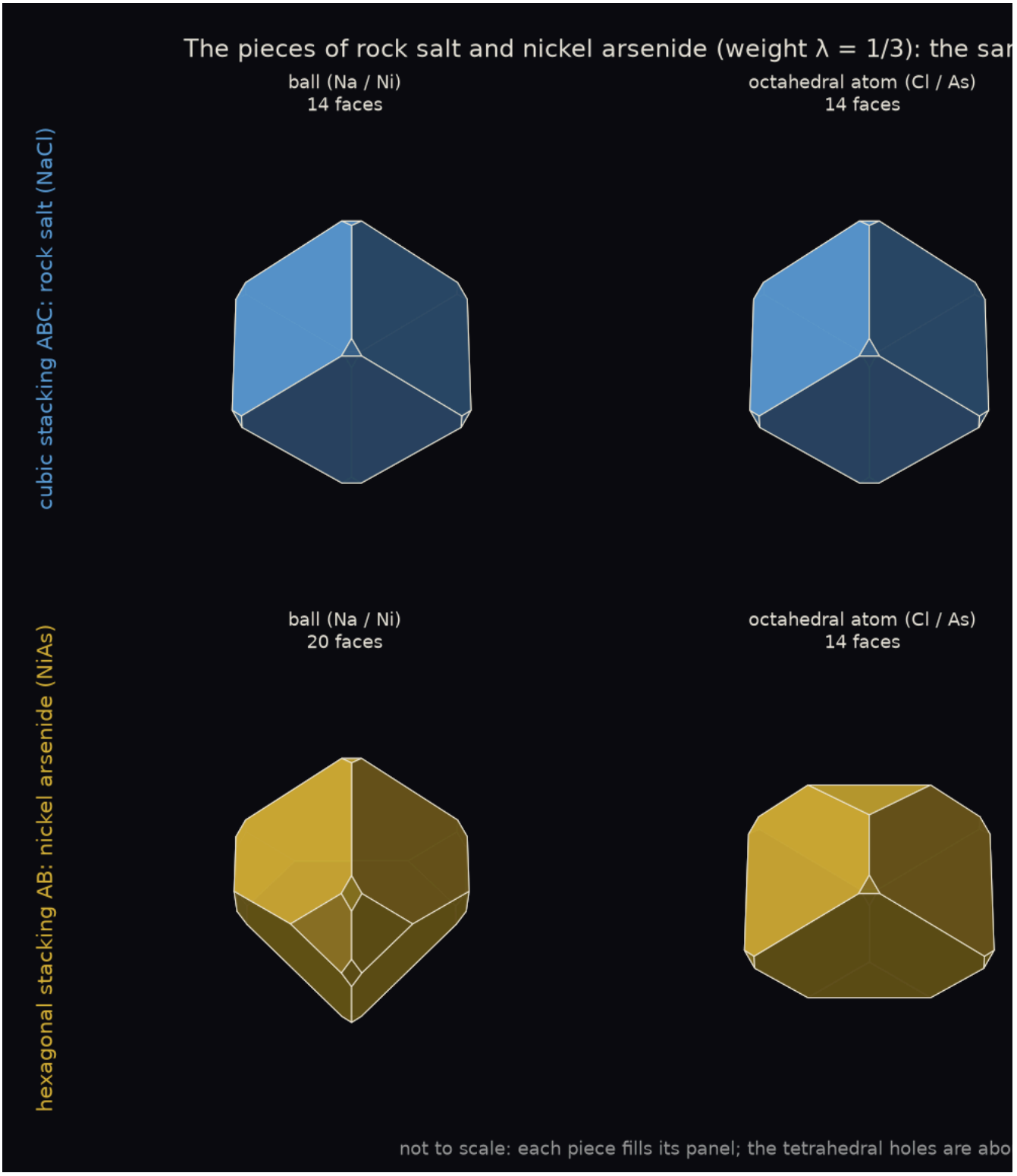


Figure 1. The pieces of rock salt (cubic stacking ABC, NaCl) and nickel arsenide (hexagonal stacking AB, NiAs) at weight λ = 1/3: the ball, the octahedral atom and a tetrahedral hole, each the power cell of its site. Their volumes agree with the exact volumes of the run to nine decimals. Not to scale: each piece fills its panel; the tetrahedral holes are in fact about a thousandth of the others. A mixed stacking is built from halves of these pieces, and by the Hexagonality Law each layer holds exactly the share of its own row.

3. Any stacking

> **Theorem 1.** Let a role satisfy the Swap Rule. For any sequence of N consecutive layers of any stacking, periodic or not, the total volume T of that role's pieces in the N − 1 gaps between them satisfies $|T - (n_c F(c, c) + n_h F(h, h))| \le 2 \cdot \max|f| + 2 \cdot \max|g|$, where n_c, n_h count the cubic and hexagonal layers. Hence, in any stacking whose share of hexagonal layers tends to h, the average volume per layer tends to F(c, c) + h·(F(h, h) − F(c, c)), whatever the order of the layers and whatever the correlations between them.

Proof. With F(x, y) = f(x) + g(y), the gaps between layers 1, …, N give $T = \Sigma_{k<N} f(t_k) + \Sigma_{k>1} g(t_k) = \Sigma_k (f + g)(t_k) - g(t_1) - f(t_N)$, and (f + g)(c) = F(c, c), (f + g)(h) = F(h, h). The two end terms are bounded; dividing by N gives the limit. ∎

So the total needs only h. The *split* of the pieces among the four kinds cc, ch, hc and hh does depend on how neighbouring layers are correlated, and the Markov descriptions of disordered stackings (Hart, Hansen and Kuhs, 2021) give it; the totals are independent of all of that.

Where h is measured. In crystals of nearly hard colloidal spheres the layers are often randomly stacked (random hexagonal close packing): each new layer takes either of the two positions not used by the layer below, with probability α for the cubic choice; α = ½ is completely random stacking, and then h = ½ (Pusey et al., 1989; later work discusses whether faulted-twinned cubic crystals are the stable form, Kegel and Dhont, 2000, and how such crystals age, Martelozzo et al., 2002). In ice frozen from supercooled water the layers are cubic and hexagonal in random order, stacking-disordered ice I_{sd} (Malkin et al., 2012), whose cubicity 1 − h is measured by diffraction: 73.3% in the most cubic bulk sample (Malkin et al., 2015), 78% in 20 nm particles (Amaya et al., 2017). The framework of ice and of diamond is the stacking of #60's pyrochlore run (zinc blende with both atoms alike), where the law held in all 96 exact tests (#66).

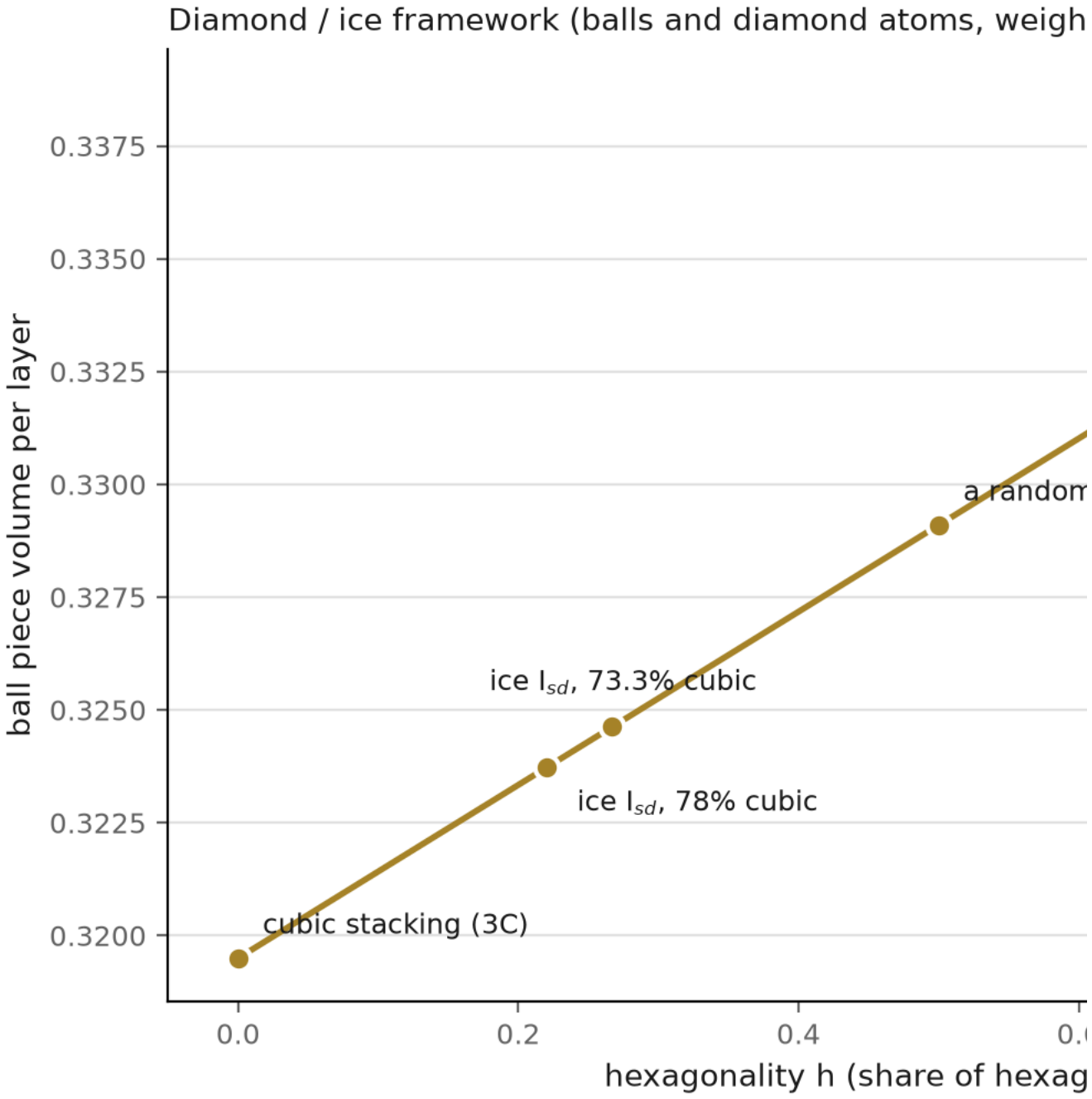


Figure 2. The diamond / ice framework (#60's pyrochlore run: balls and diamond atoms of weight λ = 1/16, holes 0): the volume of the ball pieces per layer is an exact straight line in the hexagonality h (Theorem 1). Marked: the two pure stackings, a completely random stacking (h = 1/2), and the measured cubicities of stacking-disordered ice (73.3%, Malkin et al. 2015; 78%, Amaya et al. 2017).

4. Stacking faults

> **Theorem 2.** Let a role satisfy the Swap Rule. Compared with a perfect crystal of the same number of layers, a stacking fault that changes Δ layers from cubic to hexagonal (Δ < 0: from hexagonal to cubic) changes the total volume of that role's pieces by exactly Δ·(F(h, h) − F(c, c)).

Proof. Theorem 1 with the same N and the same end layers: T changes by $(\Delta n_h)F(h, h) + (\Delta n_c)F(c, c)$ with $\Delta n_c = -\Delta n_h$. ∎

Counting the layers with Jagodzinski's rule on the classical sequences of each fault (perfect and faulted runs of 18 layers, read away from the ends):

Fault	Sequence	Layers changed
cubic crystal: twin boundary	…ABCABC**B**ACBAC…	1 becomes hexagonal
cubic crystal: intrinsic fault (a layer removed)	…ABCABC**BC**ABC…	2 become hexagonal
cubic crystal: extrinsic fault (a layer inserted)	…ABCABC**B**ABC…	2 become hexagonal
hexagonal crystal: growth fault I1	…ABAB**C**BCB…	1 becomes cubic
hexagonal crystal: deformation fault I2	…ABAB**CA**CA…	2 become cubic
hexagonal crystal: extrinsic fault E	…ABAB**C**ABAB…	3 become cubic

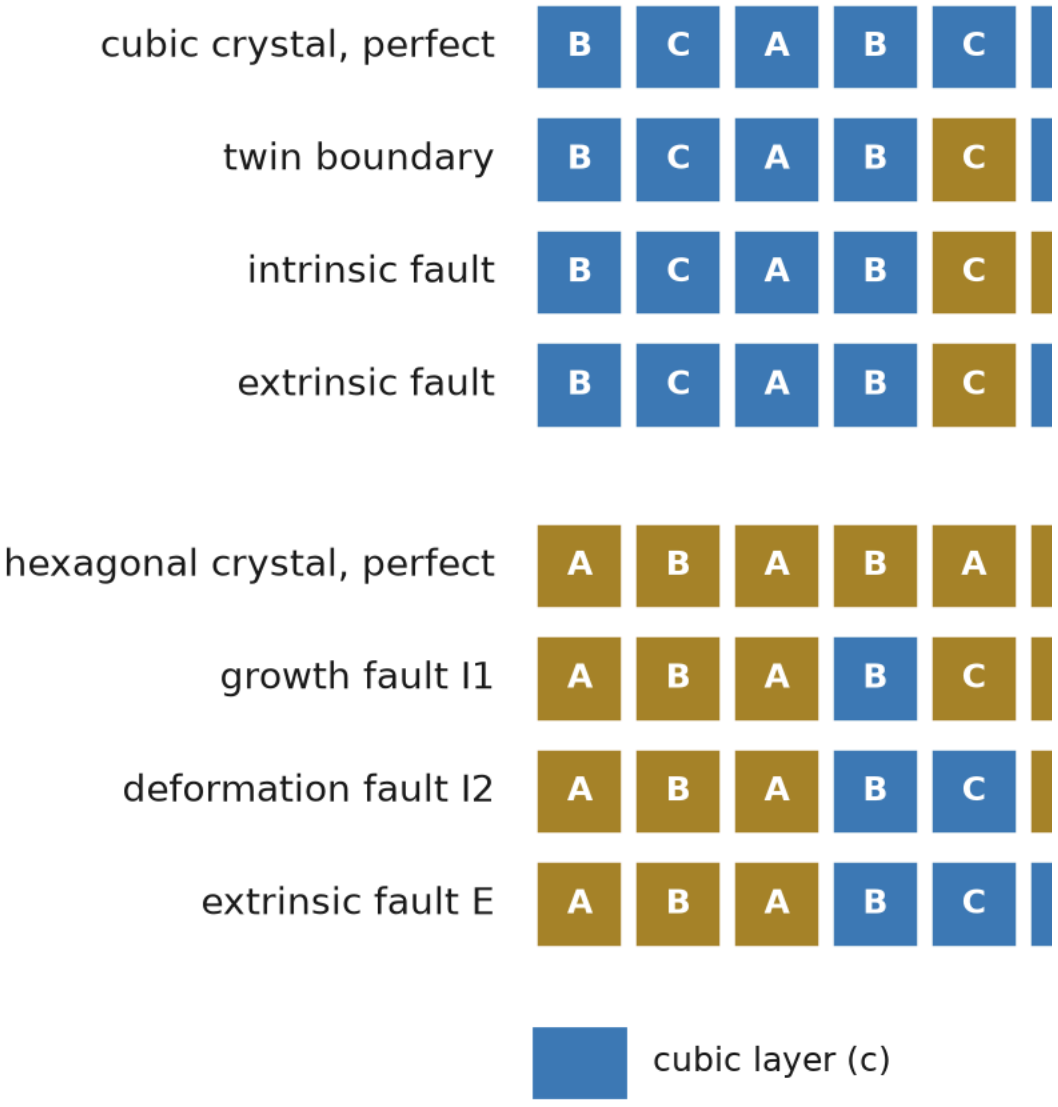


Figure 3. The six classical faults as strips of layers, each layer coloured by its type (blue cubic, gold hexagonal; Jagodzinski's rule), ten layers round the fault. A twin turns one layer of a cubic crystal hexagonal, an intrinsic or an extrinsic fault two; the faults of a hexagonal crystal turn one, two and three layers cubic.

So the faults of cubic crystals stand 1 : 2 : 2 and those of hexagonal crystals 1 : 2 : 3, in their effect on every kind of piece. In the physics of stacking faults the energies are ordered in the same way, in terms of the number of layers of the other kind that a fault creates, to lowest order (ANNNI descriptions of fault energies); here the ratios are exact, for the space each atom and hole occupies.

Rock salt. In rock salt and nickel arsenide (balls and octahedral atoms of weight λ, #64), at $\lambda = 1/3$ (in the window $7/24 < \lambda < 3/8$), each layer that a fault turns hexagonal changes the pieces by exactly:

Piece	V(cc)	V(hh)	Change per hexagonal layer
ball	0.49961	0.52016	+71/3456
octahedral atom	0.49961	0.47888	−215/10368
tetrahedral hole (lower)	0.00039	0.00048	+1/10368
tetrahedral hole (upper)	0.00039	0.00048	+1/10368

The four changes add up to exactly zero (213/10368 − 215/10368 + 2/10368): a fault moves space from the octahedral atoms to the balls and the small tetrahedral holes, and creates or loses none.

5. The Laves foot atom on longer stackings

The proof of the law needs each piece to be fixed by its two neighbouring layers. In the Laves phases (#64) the foot atom, the big atom M0, reads three layer types above $\lambda = 17/96$, so the proof does not apply to it; #66 found the law exact for it on every stacking up to six layers and left it as an open problem. Faults and random stackings bring longer runs of layers, so we extended the test to the nine further close-packed stackings of seven and eight layers (ABACACB, ABABACB, ABCBACB, ABACBACB, ABCBABAB, ABCBCBAB, ABCBCACB, ABCACBAB, ABCABACB), each walked exactly in rational arithmetic with #57's census engine (all census checks passed), and tested every role in every window of λ against $n_c \cdot T(ABC)/3 + n_h \cdot T(AB)/2$.

> **Result.** On all 13 non-pure close-packed stackings up to eight layers, in all five windows of λ, every role of the Laves phases, the foot atom M0 included, totals exactly $n_c F(c, c) + n_h F(h, h)$: 390 of 390 exact tests.

So the law reaches beyond its proof: a piece that depends on three layers still adds up layer by layer on every stacking tested. Whether it does so on every stacking remains an open problem.

6. Conclusion

Wherever the Swap Rule holds, the space of a stacking is counted layer by layer: each layer brings exactly its own pure share. A random stacking is therefore fixed, in its totals, by one number, the share of hexagonal layers, which is what diffraction measures in randomly stacked colloids and in stacking-disordered ice; and every stacking fault moves space by a whole number of layer shares, 1, 2 and 2 for the faults of cubic crystals and 1, 2 and 3 for those of hexagonal crystals.

7. Prior art and credit

Random hexagonal close packing of colloids and its stacking probability α are due to Pusey and coworkers (1989); stacking-disordered ice and its cubicity to Malkin, Murray, Salzmann and coworkers (2012, 2015); Markov descriptions of stacking disorder are reviewed by Hart, Hansen and Kuhs (2021); correlations in randomly stacked solids are studied by Ganesh and Nasr (2023). The classical fault sequences are textbook material, and the ANNNI description of fault energies in terms of polytype energies is standard. We found no statement that the volumes of the weighted pieces of a stacking are fixed exactly by its hexagonality, or that faults change them by whole multiples of one layer share. The Hexagonality Law is #66; the rule of halves is #57 (with #61 for squashed stackings and #64 for crystals), and the Swap Rule is #61 (with #64 for crystals).

> **Version 2** (28 October 2026): adds three figures (the pieces of rock salt and nickel arsenide, the straight line for the diamond / ice framework with measured ice, and the six faults as strips of layers).

> **Sources checked** (28 October 2026). *Reference library* (ask-library: "randomly stacked close packed colloidal crystals hexagonality stacking probability"; "stacking fault energy intrinsic extrinsic twin ratio fcc hcp ANNNI"; "hydrogen interstitial octahedral tetrahedral sites fcc hcp"): R. Ganesh and A. K. Nasr, Correlations in randomly stacked solids (2023, arXiv 2306.13569) and A. G. Hart, T. C. Hansen and W. F. Kuhs, A Markov theoretic description of stacking disordered aperiodic crystals including ice and opaline silica (2021, arXiv 2102.08418), catalogue abstracts; Howie and Marks (1984), Marks (1984), Hofmeister (1998) on twinned particles, not relevant. *Web* (search summaries, not full texts): Martelozzo et al., Phys. Rev. E (2002), summarising Pusey et al. (1989); Kegel and Dhont (2000); arXiv 1406.7697; Malkin et al., PNAS (2012), and later cubicities (arXiv 1812.04333; J. Appl. Cryst. 2018; Phys. Chem. Chem. Phys. 2015, c4cp02893g); ANNNI fault energies: arXiv 1704.00313, 2111.03591, 2405.04876, 1505.00431, Phys. Rev. B 56, 10844, and a density-functional study of Cu, Ag and Au (Leoben). Pusey et al. (1989) and Malkin et al. (2012) were not read in full.

> **Data and code.** Study 2026-10-28-random-stackings-faults (scripts/faults.py, faults.json, fault_volumes_rocksalt.json, scripts/hexlaw_laves8.py and its output), using the exact weight walks of study 2026-10-27-splash-rules (run2) and the law check of study 2026-10-28-halves-physics, attached to this record. References: DICTO, The Focus (#66), DICTO DOI 10.5281/zenodo.23717687; The Swap Rule in Crystals (#64), 10.5281/zenodo.23709753; DICTO Janus c/h Pyrochlore (#60), 10.5281/zenodo.23678750; The Rule of Halves (#57), 10.5281/zenodo.23647898.